

Code-Layout, Readability and Reusability

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the code quality of the Emotion Detection Learning Support Engine in terms of its structure, readability, and reusability. The project follows a modular architecture with separate components for authentication, emotion detection, recommendation generation, analytics, session history, and report generation. Meaningful naming conventions, proper code formatting, comments, and reusable functions improve maintainability, scalability, and make the application easier to extend for future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	yes	Maintained consistent 4-space indentation and proper code formatting throughout the project.
2	Proper File Structure	Files and folders are logically organized	yes	Project organized into separate folders such as authentication, analytics, database, models, utils, saved_models, data and Streamlit application.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	We used names like MODEL_PATH, tokenizer, prediction, user_input, etc.
4	Function / Method Names	Functions are descriptively named	yes	Functions like predict_emotion(), model loading functions, authentication functions are descriptive.
5	Code Comments	Inline and block comments explain logic	yes	Added comments explaining important sections including model loading, preprocessing, prediction and database operations.
6	Modular Design	Code is split into reusable functions/modules	yes	Authentication, prediction, analytics, and database are separate modules.
7	No Redundant Code	Duplicate or unused code is removed	yes	We removed duplicate code while fixing your Streamlit app

8	Error Handling	Exceptions and errors are handled gracefully	yes	Used exception handling while loading models, accessing files and connecting to the database.
---	----------------	--	-----	---

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python, Streamlit	User Login & Registration	High

2	Emotion Detection Module	Python, TensorFlow, PyTorch (BERT)	Emotion Prediction	High
3	Recommendation Engine	Python	Personalized Learning Suggestions	High
4	Report Generation Module	Python	Download Analysis Report	Medium
5	Analytics & Session History	Python, CSV, Streamlit	Python, CSV, Streamlit	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with modular folder structure and separate components.
Readability	5	Code uses meaningful variable names, proper indentation, and clear formatting.
Reusability	5	Core functionalities are implemented as reusable modules and functions.
Documentation / Comments	5	Code contains section-wise comments and is easy to understand.
Overall Score	5	The project follows good coding standards, modular design, and reusable architecture.